

Elements of Statistics and Probability

Nabil Ahmed Uthso

School of Data and Computational Sciences

BRAC University

E-mail: nabil.ahmed@bracu.ac.bd

Table of Contents

1 Measures of Central Tendency

2 Measures of Position

Measures of Central Tendency

- Measure of **Central Tendency** is a single value that summarizes a set of data
- It locates the center of the values
- Also known as **measure of location** or the **measure of average** or simply the **averages**
- In most datasets, the values show a distinct tendency to group or cluster around a central point. This tendency of clustering the values around the center of the series is usually called **central tendency**

Measures of Central Tendency

- It aims to provide an accurate description of the entire data
- It is the single value that is most typical/ representative of the collected data
- Central tendency can be considered as the first summary of the data

Different Measures of Central Tendency

- **Mean:**
 - Arithmetic Mean (AM)
 - Geometric Mean (GM)
 - Harmonic Mean (HM)
- **Median**
- **Mode**

Arithmetic Mean

- The **mean** is defined to be the sum of the data values divided by the total number of values
- We will compute two means: one for the **sample** and one for a finite **population** of values
- The mean, in most cases, is not an actual data value

Sample Mean

- The symbol $\bar{X}$ represents the sample mean
- $\bar{X}$ is read as “X-bar”
- The Greek symbol Σ is read as “sigma” and it means “to sum”
- For a set of observations $x_1, x_2, \dots, x_n$, the sample arithmetic mean is

$$\begin{aligned}\bar{X} &= \frac{x_1 + x_2 + \dots + x_n}{n} \\ &= \frac{1}{n} \sum_{i=1}^n x_i\end{aligned}$$

Sample Mean (Example)

- The marks (out of 25) obtained by 8 students in a class test are 10, 19, 12, 21, 18, 20, 11, and 19. What is the arithmetic mean of the marks obtained by the students?
- The sample mean is

$$\begin{aligned}\bar{X} &= \frac{1}{8} \sum_{i=1}^8 X_i = \frac{x_1 + x_2 + \dots + x_8}{8} \\ &= \frac{10 + 19 + 12 + 21 + 18 + 20 + 11 + 19}{8} \\ &= 16.25\end{aligned}$$

Population Mean

- The Greek symbol μ represents the population mean
- The symbol μ is read as “mu”
- N is the size of finite population
- Formula for the population arithmetic mean is

$$\begin{aligned}\mu &= \frac{x_1 + x_2 + \dots + x_N}{N} \\ &= \frac{1}{N} \sum_{i=1}^N x_i\end{aligned}$$

Population Mean (Example)

- A small company consists of the owner the manager the salesperson and two technicians The salaries are listed as \$50,000, \$20,000, \$12,000, \$9,000 and \$9,000, respectively (assume this is the population)
- Then the population mean is

$$\begin{aligned}\mu &= \frac{1}{5} \sum_{i=1}^5 X_i = \frac{x_1 + x_2 + \dots + x_5}{5} \\ &= \frac{50,000 + 20,000 + 12,000 + 9,000 + 9,000}{5} \\ &= \$20,000\end{aligned}$$

Properties of the Arithmetic Mean

- Every set of interval-level and ratio-level data has a mean
- All the values are included in computing the mean
- A set of data has a unique mean
- The arithmetic mean is the only measure of central tendency where the sum of the deviations of each value from the mean is zero
- Sum of deviations of each value from the mean is zero, i.e.,

$$\sum_i (X_i - \bar{X}) = 0$$

Sample Mean for Single-Value Grouping

- The mean for single-value grouping is given by

$$\bar{X} = \frac{\sum_i f_i x_i}{n} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{n}$$

- Here, f is the frequency for the corresponding value of x and

$$n = \sum_i f_i$$

Sample Mean for Single-Value Grouping (Example)

- The scores for students on a point quiz are given in the table.

Find the mean score

Score, X	Frequency, f
0	2
1	4
2	12
3	4
4	3

Sample Mean for Single-Value Grouping (Example)

Score, x_i	Frequency, f_i	$f_i x_i$
0	2	0
1	4	4
2	12	24
3	4	12
4	3	12
Total	$n = \sum_i f_i = 25$	$\sum_i f_i x_i = 52$

• Therefore, $\bar{X} = \frac{\sum_i f_i x_i}{n} = \frac{52}{25} = 2.08$

Sample Mean for Limit/Cut-Point Grouping

Class Interval	Class Midpoint	Frequency
$L_1 - U_1$	x_1	f_1
$L_2 - U_2$	x_2	f_2
$\vdots$	$\vdots$	$\vdots$
$L_i - U_i$	x_i	f_i
$\vdots$	$\vdots$	$\vdots$
$L_n - U_n$	x_n	f_n

Sample Mean for Limit/Cut-Point Grouping

- The mean for limit/cut-point grouping is given by

$$\bar{X} = \frac{\sum_i f_i x_i}{n} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{n}$$

- Here, f is the frequency, x is the corresponding class midpoint, and $n = \sum_i f_i$

Sample Mean for Limit/Cut-Point Grouping

- Given the table below, find the mean

Class	Frequency, f
15.5 – 20.5	3
20.5 – 25.5	5
25.5 – 30.5	4
30.5 – 35.5	3
35.5 – 40.5	2

Sample Mean for Limit/Cut-Point Grouping

Class	Midpoint, x_i	Frequency, f_i	$f_i x_i$
15.5 – 20.5	18	3	54
20.5 – 25.5	23	5	115
25.5 – 30.5	28	4	112
30.5 – 35.5	33	3	99
35.5 – 40.5	38	2	76
Total	—	$n = \sum_i f_i = 17$	$\sum_i f_i x_i = 456$

• Therefore, $\bar{X} = \frac{\sum_i f_i x_i}{n} = \frac{456}{17} = 26.82$

Advantages of Arithmetic Mean

- It is well defined, easy to understand & simple to calculate
- It includes all the observations
- It minimises error in the prediction of any one value in data set
- It provides equal weight to all the observations

Disadvantages of Arithmetic Mean

- It is affected by extreme values (outliers)
- It cannot be calculated for open-end classes
- Cannot be used in case of qualitative data

Weighted Mean

- The weighted mean is a special case of the arithmetic mean
- It occurs when there are several observations of the same value
- Instead of each data point contributing equally to the final mean, some data points contribute more weight than others

Note: If all the weights are equal, then the weighted mean equals the arithmetic mean

Weighted Mean

- The weighted mean is given by

$$\bar{X} = \frac{\sum_i w_i x_i}{\sum_i w_i} = \frac{w_1 x_1 + w_2 x_2 + \dots + w_n x_n}{w_1 + w_2 + \dots + w_n}$$

- Here, x is the particular value of data set and w is the corresponding weight

Weighted Mean (Example)

- Concord sells 3-bedroom apartment for Tk 3 millions, 2-bedroom apartment for Tk 2.5 millions, and 4-bedroom apartments for Tk 4.5 millions. Among the last 10 apartments they sold, 2 were 3-bedroom apartments, 5 were 2-bedroom apartments and 3 were 4-bedroom apartments. What is the average price of the last 10 apartments sold by Concord?

Harmonic Mean (HM)

- The Harmonic Mean (HM) is defined as the reciprocal of the average of the reciprocals of the data values
- Harmonic mean gives less weight to the large values and large weight to the small values to balance the values correctly
- In general, the harmonic mean is used when there is a necessity to give greater weight to the smaller items
- It is applied in the case of times and average rates

Harmonic Mean

- For a set of observations $x_1, x_2, \dots, x_n$; ($x_i > 0, \forall i = 1, 2, \dots, n$), the harmonic mean is defined by

$$HM = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

When to Use Harmonic Mean

- Use the harmonic mean when dealing with rates, ratios, or inversely proportional quantities
- Calculate the average speed or velocity over a given distance
- Give more weight to lower values and balance the contributions of different rates
- Avoid distortion caused by outliers or extremely high rates

Harmonic Mean (Example)

- A boat travels upstream from B to A at a speed of 15km/hr and down stream from A to B at a speed of 25km/hr . What is the average speed of the boat?
- The average speed is

$$HM = \frac{2}{\frac{1}{15} + \frac{1}{25}} = 18.75\text{km/hr}$$

Harmonic Mean (Example)

- The harmonic mean of three numbers is 10, and the first numbers is 15. The average of the first two numbers is 12. Find the other number

Geometric Mean (GM)

- Geometric mean is often used for a set of numbers whose values are meant to be multiplied together or are exponential in nature
- For example data on the growth of the human population or interest rates of a financial investment
- We conduct **arithmetic mean** when observation are **independent** to each other and perform **geometric mean** when observations are **dependent** to each other

Geometric Mean

- For a set of observations $x_1, x_2, \dots, x_n$; ($x_i > 0, \forall i = 1, 2, \dots, n$), the geometric mean is defined by

$$\begin{aligned} GM &= \sqrt[n]{\prod_{i=1}^n x_i} = \sqrt[n]{x_1 \times x_2 \times \dots \times x_n} \\ &= \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}} \end{aligned}$$

When to Use Geometric Mean

- Use the geometric mean when dealing with multiplicative relationships or exponential growth
- Calculate the average growth rate or compound annual growth rate
- Handle skewed distributions or data with outliers more effectively than the arithmetic mean
- **Note that** the geometric mean is not suitable for data that includes zero or negative values since these would result in undefined or complex values

Geometric Mean (Example)

- Suppose you own a piece of art that increases in value by 50% the first year after you buy it, 20% the second year, and 90% the third year. In such circumstances at the end of
 - first year the value was multiplied by 150% or 1.5
 - the second year the value **at the end of year 1** was multiplied by 120% or 1.2
 - at the end of the third year the value **at the end of year 2** was multiplied by 190% or 1.9

Geometric Mean (Example)

- Art work value year 0: 90,000 tk
- Art work value year 1: $90,000 \times 1.5 = 135,000$ tk
- Art work value year 2: $135,000 \times 1.2 = 162,000$ tk
- Art work value year 3: $162,000 \times 1.9 = 307,800$ tk

Geometric mean (Example)

- We can also calculate this using geometric mean
- First the geometric mean of the increasing rates

$$(1.5 \times 1.2 \times 1.9)^{(1/3)} = 1.51$$

- Here, this 1.51 means if we multiply our initial investment by 1.51 each year, we would get the same amount as if we had multiplied it by 1.5, 1.2, and 1.9
- So using geometric mean we get

$$90,000 \times 1.51 \times 1.51 \times 1.51 = 307,800 \text{ tk}$$

Relationship Between AM, GM, HM

- $AM \geq GM \geq HM$
- $GM^2 = AM \times HM$

Exercise

- The harmonic mean of three numbers is 10, and the first numbers is 15. The average of the first two numbers is 12. Find the geometric mean

Median

- **Median:** The midpoint of the values after they have been ordered from the smallest to the largest, or the largest to the smallest.
- There are as many values above the median as below it in the data array
- It is the central value of the variable, when the values are arranged in order of magnitude

How to Calculate Median

- For a set of observations $x_1, x_2, \dots, x_n$ the following is the formula for the Median
 - When n is **odd**
 - Order the observations
 - Median = $\left(\frac{n+1}{2}\right)^{th}$ observation
 - When n is **even**
 - Order the observations
 - Median = average of $\left(\frac{n}{2}\right)^{th}$ and $\left(\frac{n}{2} + 1\right)^{th}$ observations

Median (Example)

- Find the median of the scores (out of 100) obtained by seven students: 72, 56, 34, 99, 89, 81, 10
- Solution:
 - Here, $n = 7$ (odd)
 - Ordered observations: 10, 34, 56, **72**, 81, 89, 99
 - Median = $\left(\frac{n+1}{2}\right)^{th} = \left(\frac{7+1}{2}\right)^{th} = 4^{th} \text{ observation} = 72$

Median (Example)

- The ages of a family of eight members are given as 12, 7, 2, 34, 17, 40, 21 and 19. Find the median age
- Solution:
 - Here, $n = 8$ (even). Ordered data: 2, 7, 12, **17**, **19**, 21, 34, 40

$$\begin{aligned}\text{Median} &= \text{average of } \left(\frac{n}{2}\right)^{th} \text{ and } \left(\frac{n}{2} + 1\right)^{th} \text{ observations} \\ &= \text{average of } \left(\frac{n}{2}\right)^{th} \text{ and } \left(\frac{n}{2} + 1\right)^{th} \text{ observations} \\ &= \frac{1}{2}(17 + 19) = 18\end{aligned}$$

- So the Median Age of the family is 18 years

Median for Grouped Data

- **Median Class:** The median class is the class interval whose cumulative frequency is greater than and nearest to $\frac{n}{2}$
- Median for grouped data is calculated by

$$M_e = L_0 + \frac{\frac{n}{2} - F_{-1}}{f} \times h$$

- L_0 = Lower limit of the median class
- n = Number of total observations
- F_{-1} = Cumulative frequency of the pre-median class
- f = Frequency of the median class
- h = Class interval of median class

Median for Grouped Data (Example)

Height (cm)	Frequency	Cumulative Frequency
60 – 63	5	5
63 – 66	18	23 $\rightarrow F_{-1}$
66 – 69	42 $\rightarrow f$	65
69 – 72	27	92
72 – 75	8	100

- Here, $\frac{n}{2} = \frac{100}{2} = 50$. So, the 50th ordered observation is in class 66 – 69, which is the median class

Median for Grouped Data (Example)

- Therefore, the median will be

$$\begin{aligned}M_e &= L_0 + \frac{\frac{n}{2} - F_{-1}}{f} \times h \\&= 66 + \frac{\frac{100}{2} - 23}{42} \times 3 \\&= 67.93\end{aligned}$$

Properties of Median

- There is a unique median for each data set
- It is not affected by extremely large or small values and is therefore a valuable measure of central tendency when such values occur
- It can be computed for ratio-level, interval-level, and ordinal-level data
- It can be computed for an open-ended frequency distribution if the median does not lie in an open-ended class

Advantages of Median

- Median can be calculated in all distributions
- Median can be ascertained even with the extreme items
- It can be located graphically
- It can be determined for ratio, interval, and ordinal scale

Disadvantages of Median

- It is not based on all the values
- It is not capable of further mathematical treatment
- It is affected by fluctuation of sampling
- In case of even number of values it may not a value from the data

Usages of Median

- Whenever a data set has extreme values, the median is the preferred measure of central location
- The median is the measure of location most often reported for annual income and property value data because few extremely large incomes or property values can inflate the mean

Mode

- The **mode** is the value of the observation that appears most frequently
- It is useful measure of central tendency for nominal and ordinal level of measurements
- It can be computed for all levels of data
- Data set may have no mode because no value is observed more than once
- For ungrouped data/categorical variable, mode is the value of the variable for which the frequency is highest

Mode (Example)

- The exam scores for ten students are: 81, 93, 84, 75, 68, 87, 81, 75, 81, 87. Since the score of 81 occurs the most, the modal score is 81

Mode for Grouped Data

- **Modal Class:** The modal class is the class interval with the largest frequency
- The formula used to calculate the mode from grouped data is:

$$Mode = L_0 + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h$$

- L_0 = Lower limit of the modal class
- Δ_1 = Difference between frequencies of modal and pre-modal class = $f_0 - f_{-1}$
- Δ_2 = difference between frequencies of modal and post-modal class = $f_0 - f_1$
- h = Class interval of the modal class

Mode for Grouped Data (Example)

- Calculate the mode for the following frequency distribution

Height (cm)	Frequency
60 – 63	5
63 – 66	18 $\rightarrow f_{-1}$
66 – 69	42 $\rightarrow f_0$
69 – 72	27 $\rightarrow f_1$
72 – 75	8

Mode for Grouped Data (Example)

- Solution:

$$\begin{aligned} \text{Mode} &= L_0 + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h \\ &= 66 + \frac{(42 - 18)}{(42 - 18) + (42 - 27)} \times 3 \\ &= 67.84 \end{aligned}$$

Properties of Mode

- Unlike the mean and median, **mode is not unique**; that is, there may be more than one mode for a set of data
- If the data have exactly **two modes**, the data are **bimodal**
- If the data have **more than two modes**, the data are **multimodal**

Advantages of Mode

- Mode is readily comprehensible and easily calculated
- It is not at all affected by extreme value
- It can be determined for any scales of measurement
- The value of mode can also be determined graphically

Disadvantages of Mode

- It is not based on all observations
- It is not capable of further mathematical treatment
- Mode is affected to a great extent by sampling fluctuations
- Choice of grouping has great influence on the value of mode

Table of Contents

1 Measures of Central Tendency

2 Measures of Position

Quartiles

- **Step 1:** Arrange the data in increasing order
- **Step 2:** Find the median of the entire data set. This value is the second quartile, Q_2
- **Step 3:** Divide the ordered data set into two halves, a bottom half and a top half; if the number of observations is odd, include the median in both halves
- **Step 4:** Find the median of the bottom half of the data set. This value is the first quartile, Q_1
- **Step 5:** Find the median of the top half of the data set. This value is the third quartile, Q_3

Quartiles (Example)

- From Tropical Cyclone reports, published by the National Hurricane Center, we obtained the data shown in the table on maximum wind speeds, in miles per hour (mph), for one years tropical cyclones in the Atlantic Basin. Determine and interpret the quartiles for these data.

60	70	85	65	100	60	110	45	80	40
105	80	115	90	50	45	90	115	50	

Quartiles (Example)

- **Solution:** In summary, the three quartiles for the maximum wind speeds in the table are $Q_1 = 55$ mph, $Q_2 = 80$ mph, and $Q_3 = 95$ mph
- **Interpretation:** We see that roughly 25% of the maximum wind speeds are less than 55 mph, roughly 25% are between 55 mph and 80 mph, roughly 25% are between 80 mph and 95 mph, and roughly 25% are greater than 95 mph

Quartiles from Raw Data

- Calculate $k = \frac{i \times n}{4}$, $i = 1, 2, 3$
- If k is a whole number,

$Q_i = \text{average of } k^{\text{th}} \text{ and } (k + 1)^{\text{th}} \text{ observation}$

- If k is a fraction,

$Q_i = \text{round up } k \text{ to the next whole number}$

Deciles from Raw Data

- Calculate $k = \frac{i \times n}{10}$, $i = 1, 2, \dots, 9$
- If k is a whole number,

$D_i =$ average of k^{th} and $(k + 1)^{th}$ observation

- If k is a fraction,

$D_i =$ round up k to the next whole number

Percentiles from Raw Data

- Calculate $k = \frac{i \times n}{100}$, $i = 1, 2, \dots, 99$
- If k is a whole number,

$P_i =$ average of k^{th} and $(k + 1)^{th}$ observation

- If k is a fraction,

$P_i =$ round up k to the next whole number

Percentiles Corresponding to given Values

- The percentile corresponding to a given value (X) is computed by using the formula:

$$\textit{Percentile} = \frac{\text{number of values below } X + 0.5}{n} \times 100$$

Percentiles Corresponding to given Values

- A teacher gives a 20-point test to 10 students. Find the percentile rank of a score of 12. Scores: 18, 15, 12, 6, 8, 2, 3, 5, 20, 10.
- **Ordered set:** 2, 3, 5, 6, 8, 10, 12, 15, 18, 20
- $Percentile = \frac{6 + 0.5}{10} \times 100 = 65^{th}$ percentile
- **Interpretation:** The student with score 12 did better than 65% of the class

Quartiles from Grouped Data

- Q_i **Class**: The Q_i class is the class interval whose cumulative frequency is greater than and nearest to $\frac{i \times n}{4}$, $i = 1, 2, 3$
- Formula for i^{th} quartile:

$$Q_i = L_0 + \frac{\frac{i \times n}{4} - F_{-1}}{f} \times h$$

- L_0 = Lower limit of the Q_i class
- n = Number of total observations
- F_{-1} = Cumulative frequency of the pre- Q_i class
- f = Frequency of the Q_i class
- h = Class interval of Q_i class

Deciles from Grouped Data

- **D_i Class:** The D_i class is the class interval whose cumulative frequency is greater than and nearest to $\frac{i \times n}{10}$, $i = 1, 2, \dots, 9$
- Formula for i^{th} decile:

$$D_i = L_0 + \frac{\frac{i \times n}{10} - F_{-1}}{f} \times h$$

- L_0 = Lower limit of the D_i class
- n = Number of total observations
- F_{-1} = Cumulative frequency of the pre- D_i class
- f = Frequency of the D_i class
- h = Class interval of D_i class

Percentiles from Grouped Data

- P_i **Class**: The P_i class is the class interval whose cumulative frequency is greater than and nearest to $\frac{i \times n}{100}$, $i = 1, 2, \dots, 99$
- Formula for i^{th} percentile:

$$P_i = L_0 + \frac{\frac{i \times n}{100} - F_{-1}}{f} \times h$$

- L_0 = Lower limit of the P_i class
- n = Number of total observations
- F_{-1} = Cumulative frequency of the pre- P_i class
- f = Frequency of the P_i class
- h = Class interval of P_i class

Exercise 1

- Consider the marks for 20 students in first quiz:

20	5	8	13	10	7	11	19	17	18
8	5	18	16	12	18	15	7	6	16

- (a) Find the mean, median, and mode of the dataset. Interpret the median value.
- (b) What is the percentile rank of the score 13? Interpret your findings.
- (b) Calculate the first quartile. What does this value mean? Show that $Q_1 = P_{25}$.

Exercise 2

- The provided data presents utility costs (in BDT) for a sample of 70 two-bedroom flats:

Utility cost:	1000–1500	1500–2000	2000–2500	2500–3000	3000–3500	3500–4000
Frequency:	10	12	15	17	9	7

- Calculate the arithmetic mean, median, and mode. Interpret the results.
- Calculate the third quartile (Q_3). Prove that $D_4 = P_{40}$.

Thank You